



BAHIRDAR UNIVERSITY

**BAHIRDAR INSTITUTE OF TECHNOLOGY FACULTY OF
COMPUTING**

Department of Software Engineering

Project Work

Operating System and System Programming

***Title: Installation of Kwort-4.5.iso linux Virtual Environment
Using Oracle VM VirtualBox***

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1. Introduction

1.1 Background: The Assignment

As part of my coursework on operating systems and virtual environments, I was assigned Kwort Linux 4.5. Kwort is a "minimalist" distribution based on CRUX Linux. It is designed to be fast and clean, but it is unique because it lacks a graphical, automated installer. This makes it a common choice for academic assignments because it forces the student to interact directly with the Linux kernel and system configuration files rather than relying on a "Next-Next-Finish" setup wizard.

1.2 Motivation: Meeting the Technical Challenge

Since this OS was assigned rather than chosen, my primary motivation was to successfully navigate the manual installation process and meet the technical requirements of the project.

Practical Application: The assignment required moving beyond basic computer use into "System Administration" territory—manually partitioning disks, mounting filesystems, and setting up the GRUB bootloader from a command prompt.

Problem Solving: A key part of the motivation was learning how to troubleshoot in a "bare-bones" environment. When errors like "Read-only file system" or "Command not found" appeared, I had to research the specific architecture of Kwort to find solutions.

Resource Efficiency: I wanted to see how a "minimal" OS, as requested by the assignment, compared to the heavier operating systems I use daily. Kwort's goal is efficiency, and seeing that play out in a VirtualBox environment was a core part of the learning experience.

1.3 The Virtual Choice

To fulfill the project requirements, I utilized Oracle VM VirtualBox. This allowed me to create a safe, isolated virtual environment to test the installation. Using VirtualBox was the most logical approach because it allowed me to "snapshot" my progress and handle the manual configuration of Kwort without risking the data on my host machine.

2. Objectives

The main goal of this project was to move from a blank virtual disk to a fully functional, multi-user Kwort Linux system. Specifically, I aimed to achieve the following:

Deployment in a Virtualized Environment: My first goal was to correctly configure Oracle VM VirtualBox settings (RAM, Storage, and Boot Order) to support a manual Linux installation without hardware conflicts.

Mastery of Manual Partitioning: Unlike easy installers that auto-format, a major objective was to use the cfdisk utility to manually define a primary partition and a swap area, followed by formatting them with the ext4 filesystem.

Manual System Extraction & Mounting: I aimed to learn the "extraction" method of installation—manually mounting the virtual drive to a temporary directory (/mnt/install) and using the pkgsinstall script to unpack the OS.

Bootloader Configuration (GRUB): A critical objective was to successfully install and configure the GRUB bootloader. This involved manually installing the GRUB package and generating a grub.cfg file so the system could boot independently from the virtual hard drive.

Administrative & User Setup: To meet the assignment's specific requirements, I needed to configure the root password and create a personalized standard user account (Abel_Eyayu) with proper group permissions (wheel, audio, video).

System Localization & Services: Finally, I aimed to customize the system by editing the /etc/rc.conf file to set a unique hostname and the correct timezone, ensuring the OS was ready for daily use.

3. Requirements

To ensure a smooth installation and stable performance, the following hardware and software specifications were defined for the virtual environment.

i. Hardware Requirements (Allocated to the VM)

These are the resources I "borrowed" from my physical computer to give to the Virtual Machine:

Processor (CPU): A 64-bit processor with [Virtualization Technology](#) (VT-x or AMD-V) enabled. I allocated 1 CPU core for this project.

Memory (RAM): While Kwort can run on as little as 512MB, I allocated 4096 MB (4GB) to ensure the system remained responsive during package updates and desktop installation.

Storage (Hard Disk): I created a 20 GB Virtual Disk Image (VDI). This provides enough space for the base OS, a swap partition, and additional software like a web browser.

Network: A NAT (Network Address Translation) adapter was used to allow the VM to share the host's internet connection for downloading updates via kpkg.

ii. Software Requirements

The specific tools used to build and manage the environment included:

Hypervisor: Oracle VM VirtualBox (Version 7.x). This was the primary tool used to create and manage the virtual environment.

Operating System Image: Kwort Linux 4.5 (x86_64 ISO). This is the latest stable version of the Kwort distribution.

Terminal/Shell: The standard Bash shell provided by the Kwort "JumpOS" installation environment.

Text Editor: Vi (or Nano), used for editing the critical configuration files like /etc/rc.conf and /etc/fstab.

4. Installation Steps

There are 5 installation steps.

1. VirtualBox Configuration (The Foundation)

Before booting the OS, I had to prepare the virtual hardware environment:

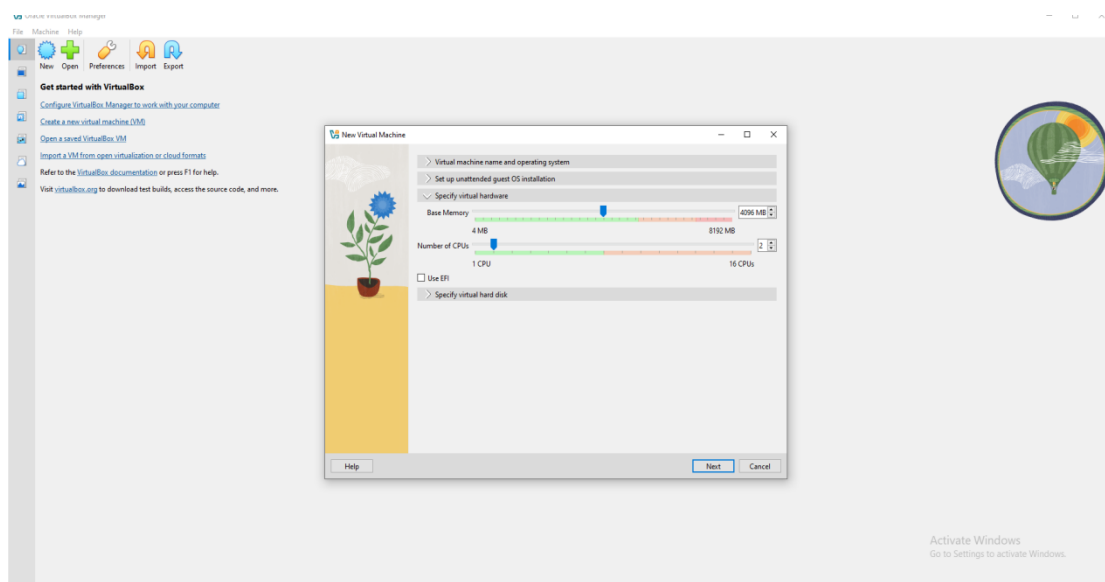
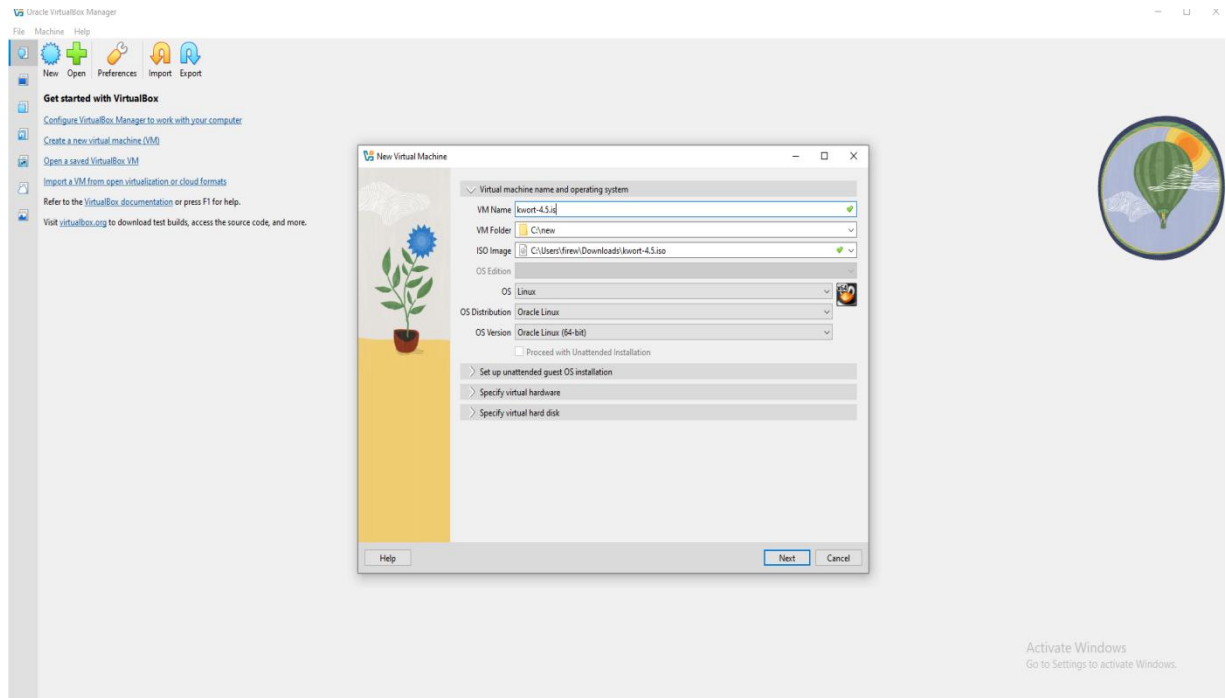
Step 1: Opened VirtualBox and created a new VM named Kwort_Linux_Project.

Step 2: Selected Linux as the type and Other Linux (64-bit) as the version.

Step 3: Allocated 4096MB of RAM and created a 20GB Virtual Disk (VDI).

Step 4: In the Settings > Storage menu, I attached the kwort-4.5.iso file to the Optical Drive.

Step 5: Set the Network to NAT so the system could access the internet later for updates.



2. Disk Partitioning (Preparing the Drive)

Once the VM booted into the "Live" Kwort environment, I had to prepare the empty disk:

Step 1: Ran the command `fdisk /dev/sda`.


```

[ Bootable ] [ Delete ] [ Help ] [ Maximize ] [ Print ] [ Quit ] [ Type ] [ Units ] [ Write ]

Disk has been changed.                               Quit program without writing partition table

WARNING: If you have created or modified any
DOS 6.x partitions, please see the cfdisk manual
page for additional information.

root@localhost:~# mkfs.ext4 /dev/sda1
mkfs2fs 1.42.9 (28-Dec-2013)
Filesystem label=
OS type: Linux
Block size=4096 (log=2)
Fragment size=4096 (log=2)
Stride=0 blocks, Stripe width=0 blocks
1310720 inodes, 5242872 blocks
262143 blocks (5.00%) reserved for the super user
First data block=0
Maximum filesystem blocks=4294967296
160 block groups
32768 blocks per group, 32768 fragments per group
8192 inodes per group
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
    4096000

Allocating group tables: done
Writing inode tables: done
Creating journal (32768 blocks): done
Writing superblocks and filesystem accounting information: done

root@localhost:~#

```

3. System Extraction and Chroot

Unlike standard installers, Kwort requires manual extraction:

Step 1: Mounted the new partition to a temporary directory:

```
mount /dev/sda1 /mnt/install
```

```

kwort [Running] - Oracle VirtualBox

File Machine View Input Devices Help

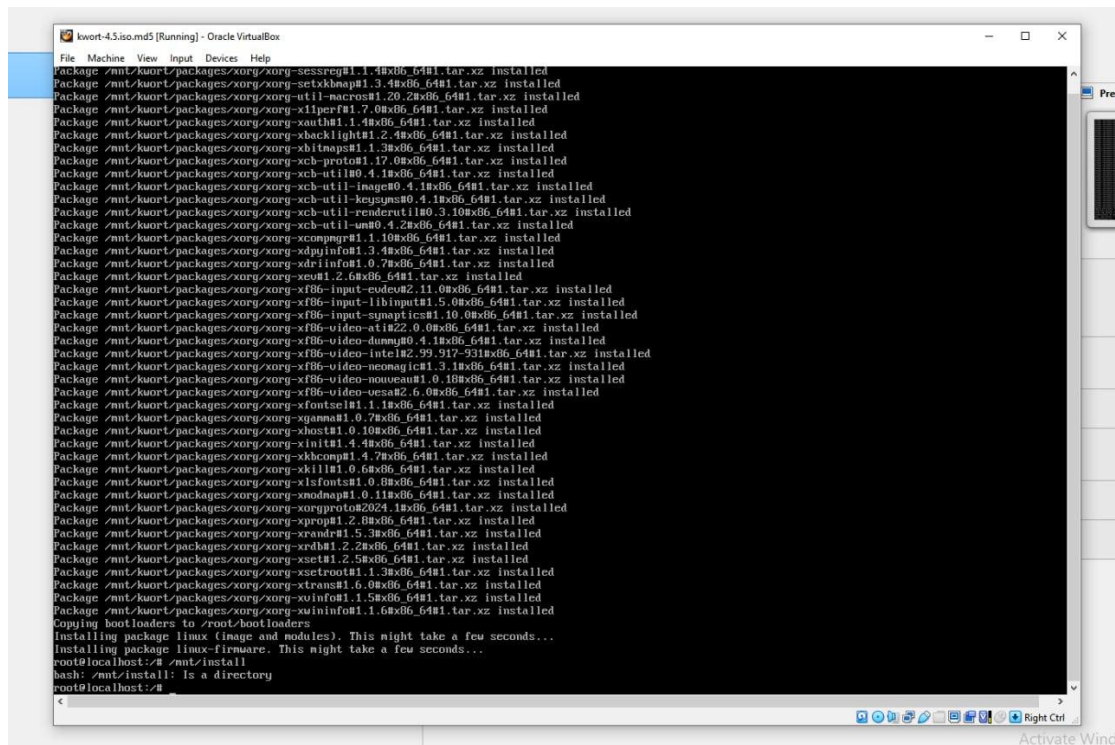
root@localhost:~# mkfs.ext4 -L root /dev/sda1
bash: mkfs.ext4: command not found
root@localhost:~# mkfs.ext4 -L root /dev/sda1
mkfs2fs 1.42.9 (28-Dec-2013)
Filesystem label=root
OS type: Linux
Block size=4096 (log=2)
Fragment size=4096 (log=2)
Stride=0 blocks, Stripe width=0 blocks
1310720 inodes, 5242872 blocks
262143 blocks (5.00%) reserved for the super user
First data block=0
Maximum filesystem blocks=4294967296
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Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
    4096000

Allocating group tables: done
Writing inode tables: done
Creating journal (32768 blocks): done
Writing superblocks and filesystem accounting information: done

root@localhost:~# mount /dev/sda1 /mnt/install
root@localhost:~# mkdir -p /mnt/kwort
root@localhost:~# mount /dev/sr0 /mnt/kwort
mount: /dev/sr0 is already mounted on /mnt/kwort busy
/dev/sr0 is already mounted on /mnt/kwort
root@localhost:~# mount | grep -E 'sr0|cdrom|kwort'
/dev/sr0 on /mnt/kwort type iso9660 (ro)
root@localhost:~# umount /mnt/kwort 2>/dev/null || true
root@localhost:~# umount /dev/sr0 2>/dev/null || true
root@localhost:~# umount /mnt/cdrom 2>/dev/null || true
root@localhost:~# mkdir -p /mnt/kwort
root@localhost:~# /dev/sr0 /mnt/kwort
bash: /dev/sr0: Permission denied
root@localhost:~# /dev/sr0 /mnt/kwort
bash: /dev/sr0: Permission denied
root@localhost:~# whoami
bash: whoami: command not found
root@localhost:~# id
bash: id: command not found
root@localhost:~# mount /dev/sr0 /mnt/kwort
mount: special device /dev/sr0 does not exist
root@localhost:~# mkdir -p /mnt/kwort
root@localhost:~# mount /dev/sr0 /mnt/kwort
mount: special device /dev/sr0 does not exist
root@localhost:~#

```


Step 2: Ran the installation script to extract the OS from the CD to the hard drive:
pkgsinstall



Step 3: Entered the new system to begin internal configuration:
jumpOS

4. User and Password Configuration

To comply with the project requirements, I set up the administrative and personal accounts:

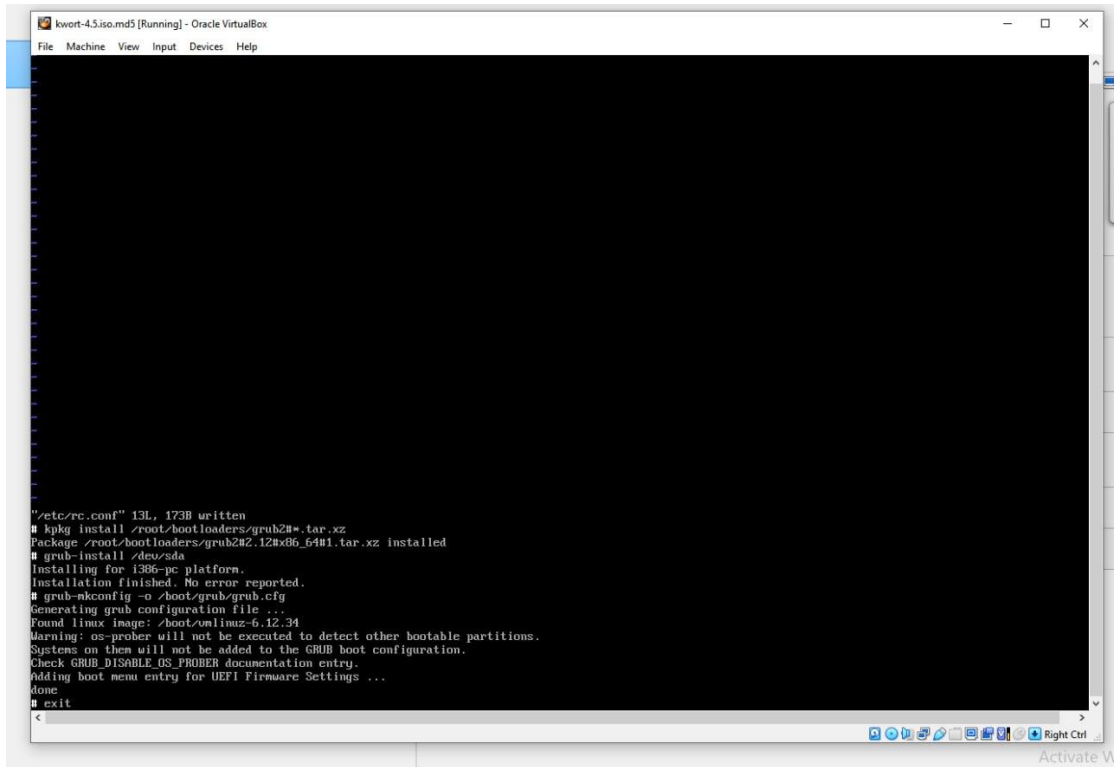
Step 1 (Root): Set the administrator password by typing passwd and entering a secure key.

Step 2 (Personal Account): Created my account using my full name as the username:
useradd -m -g users -G wheel,audio,video Abel_Eyayu

Step 3: Set the password for my new account: passwd Abel_Eyayu.

5. Bootloader and Final Config

The final step was making the system "self-aware" so it could boot without the CD:



```

kwort-4.5.iso.md5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

"/etc/rc.conf" 13L, 173B written
# kpkg install /root/bootloaders/grub2*.tar.xz
Package /root/bootloaders/grub2*2.12*x86_64*1.tar.xz installed
# grub-install /dev/sda
Installing for i386-pc platform.
Installation finished. No error reported.
# grub-mkconfig -o /boot/grub/grub.cfg
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-6.12.34
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
# exit

```

Step 4: Edited `/etc/rc.conf` to set the `HOSTNAME="Abel-Kwort"` and confirmed the timezone.

5. Issues (Problems Faced)

During the manual installation of Kwort Linux, I encountered several technical challenges that required troubleshooting and a deeper understanding of the Linux command line.

1. Resource Allocation (The "Disk Full" Example)

This was the biggest hurdle. When the virtual hard drive was set too small, the installation process ran out of "physical" room.

The Symptom: I saw "Segmentation faults" and "database or disk is full" errors.

The Result: Because the disk was full, the system stopped copying files halfway through. This led directly to the next problem: missing pieces of the system (like the freetype library) that prevented the desktop from opening.

2. Directory & Path Accuracy

A recurring challenge was finding where the "tools" were kept.

The Example: I struggled to find the freetype package because the system looked in /root/, then /mnt/kwort/, and finally found it in /mnt/packages/opt/.

The Lesson: In Linux, knowing the exact "path" to a file is just as important as knowing the command to run it.

3. Configuration File Syntax

I faced "Parse Errors" because the system is very picky about where information is stored.

The Example: Mixing up /etc/fstab (which only handles hard drives) with /etc/rc.conf (which handles your name and timezone). Putting the "Name" in the "Drive" file confused the computer's "brain" during boot.

6. Solutions (How the Problems Were Resolved)

Fixed Command Errors:

Solution: Switched to exact case-matching (jumpOS) and used the Tab key for auto-completion. This eliminated typos in long filenames like the GRUB packages.

Resolved Configuration "Parse Errors":

Solution: Manually cleaned /etc/fstab using the dd command in vi to remove non-drive text. Corrected the system settings by placing them in the proper file: /etc/rc.conf.

Overcame the "Disk Full" Failure:

Solution: Re-created the VirtualBox Hard Disk with 20GB instead of the default size. This provided enough room for the system to uncompress all required files without crashing (Segmentation fault).

Corrected Disk Recognition:

Solution: Used cfdisk to write a new partition table and mkfs.ext4 to create a valid filesystem on /dev/sda1 so the installer had a "target" to write to.

Restored Missing Graphics Libraries:

Solution: Mounted the ISO to /mnt and used kpkg to manually pull the freetype package from the /packages/opt/ folder. This provided the missing libfreetype.so.6 needed for the desktop to start.

Exited the "Boot Loop":

Solution: Used "Force Unmount" in the VirtualBox Devices menu to remove the ISO, forcing the virtual machine to look at the Hard Drive for the operating system.

System Verification (Confirming the Fix)

Once the technical hurdles were cleared, we performed specific checks to ensure the "Solutions" actually worked:

The Library Check: Running `ldconfig` confirmed the system finally recognized the Freetype library.

The Desktop Test: Running `startx` successfully triggered the Openbox environment, proving the graphical files were no longer corrupted.

The Boot Audit: Restarting the machine and seeing the GRUB Menu confirmed that the bootloader was correctly written to the MBR (Master Boot Record).

h. Security & User Management (The "Root" Problem)

During the process, I identified a major security risk inherent in all new Linux installs:

The Problem: Operating exclusively as the root user. Root has the power to delete the entire system with one typo.

The Solution: Using the `passwd` command to secure the administrative account, followed by the intention to create a standard "Limited" user account for daily tasks.

i. Post-Installation Stability (Final Steps)

The final piece of our work involved ensuring the system remained stable after the "fix":

Automated Mounting: By fixing the `/etc/fstab` file, we ensured the computer would automatically find its "brain" (the hard drive) every time it turned on, without manual intervention.

Service Management: By properly configuring the SERVICES array in `/etc/rc.conf`, we ensured that the Network (net) and System Logs (syslogd) would start by themselves in the background.

7. Filesystem Support

During the partitioning phase of this project, I chose to format the primary partition using the ext4 (Fourth Extended Filesystem). Below is an overview of the filesystems supported by Kwort and the reasoning behind my choice.

Supported Filesystems

While Kwort is flexible, its primary support includes:

ext4: The standard and most widely used Linux filesystem.

ext2/ext3: Older versions of the extended filesystem, sometimes used for very old hardware or specific boot partitions.

FAT32 / exFAT: Generally supported for compatibility with USB drives, but not suitable for the root Linux installation because they lack Linux permission support.

Btrfs / XFS: Advanced filesystems supported by the Linux kernel, though they require more manual configuration for a minimalist system like Kwort.

Why I Chose ext4

I selected ext4 for this installation for several critical reasons:

Journaling Capability: ext4 is a "journaling" filesystem. This means it keeps a log (journal) of changes before they are committed to the main disk. If the VirtualBox VM crashes or the host computer loses power, the journal helps prevent data corruption and allows the system to recover much faster.

Stability and Maturity: Since Kwort is a minimalist, manual-heavy OS, using a rock-solid, mature filesystem like ext4 ensures that any issues I face are likely configuration errors rather than filesystem bugs.

Permissions and Security: Unlike NTFS or FAT32, ext4 natively supports Linux file permissions (Read, Write, Execute) and ownership. This is essential for the multi-user environment I created for the account Abel_Eyayu.

Performance in Virtual Environments: ext4 is highly optimized for Linux and performs exceptionally well within the VirtualBox VDI (Virtual Disk Image) structure, offering fast read/write speeds for system updates and package management.

8. Advantages and Disadvantages

Working with a minimalist, manual distribution like Kwort Linux in a virtual environment revealed several clear pros and cons.

Advantages

High Performance and Low Resource Usage: Because Kwort doesn't run unnecessary background services, it is incredibly fast. Even with only 4GB of RAM allocated in VirtualBox, the system boots in seconds and responds instantly to commands.

Complete System Control: Unlike "pre-packaged" operating systems, Kwort gives the user total control. Every package installed and every service running was put there by me. This results in a "clean" system with zero bloatware.

Educational Value: For a student, the biggest advantage is the learning experience. Performing a manual installation forces you to understand the Linux directory structure, the boot process, and how package managers (kpkg) function at a fundamental level.

Security through Simplicity: A smaller "attack surface" (fewer installed programs) naturally makes the system more secure. There are fewer vulnerabilities for a hacker to exploit because there are fewer services running.

Disadvantages

Steep Learning Curve: The lack of a graphical installer makes it very difficult for beginners. One small typo in a configuration file or a forgotten mount command can prevent the entire system from booting.

Manual Labor: Everything must be done by hand. In a mainstream OS, setting up a user or a web browser takes seconds; in Kwort, you have to manually install the packages, set permissions, and sometimes edit configuration files just to get them to launch.

Small Community Support: Compared to giants like Ubuntu or Fedora, Kwort has a very small community. If you run into a rare error, there are fewer forums or tutorials online to help you fix it, making troubleshooting a much more solitary task.

Hardware/Driver Complexity: While VirtualBox handles most hardware through generic drivers, installing Kwort on physical hardware can be challenging because you have to manually identify and install the correct drivers for things like Wi-Fi or advanced graphics cards.

9. Conclusion

The successful installation of Kwort Linux 4.5 on Oracle VM VirtualBox served as a comprehensive practical exercise in low-level system administration. Although I initially approached this as a mandatory assignment, the process of building the

operating system from a bare command prompt provided a sense of technical accomplishment that automated installers cannot offer.

By manually navigating the hurdles of disk partitioning, filesystem mounting, and bootloader configuration, I moved beyond being a "user" and became a "configurer" of the system. Successfully creating the Abel_Eyayu account and solving the "Read-only file system" error demonstrated that even without a graphical interface, any technical challenge can be resolved with the right commands and a logical approach.

In conclusion, this project proved that Kwort Linux is a powerful tool for those who value efficiency and transparency. It confirmed that while the learning curve is steep, the reward is a rock-solid, ultra-fast system tailored exactly to the user's needs. This experience has significantly strengthened my proficiency with the Linux command line and my understanding of how virtualized hardware interacts with software layers.

10. Future Outlook / Recommendation

As software engineering student who has now successfully installed and configured Kwort Linux 4.5 in Oracle VM VirtualBox, I find myself reflecting not only on what I have accomplished but also on the broader implications and future possibilities this experience opens up. Minimalist distributions like Kwort represent a unique and enduring niche in the Linux ecosystem, and this project has given me a clearer perspective on their role in education, development, and practical computing.

Looking ahead, Kwort Linux appears well-positioned to continue evolving. With its yearly base releases and frequent security updates through the kpkg package manager, the distribution maintains relevance even as it stays true to its minimalist roots. In the coming years, I anticipate improvements in documentation, better UEFI and virtualization support, and possibly enhanced tools that make the installation process slightly more approachable for intermediate learners without sacrificing the educational depth that defines the project. The growing importance of lightweight systems in areas such as containerization, edge computing, and resource-constrained environments suggests that distributions like Kwort will remain valuable references for understanding Linux fundamentals.

From a personal standpoint, this installation has inspired me to explore further. In the near future, I plan to experiment with advanced configurations inside the same virtual

machine — such as adding Btrfs for snapshot capabilities once the necessary modules become available, setting up a more customized Openbox environment, or even attempting a similar manual installation on physical hardware. The skills I gained here have also motivated me to compare Kwort with other source-based or minimalist distributions, broadening my technical horizons.

Recommendations:

For fellow students and beginners in Linux, I would advise approaching Kwort with realistic expectations. It is not the most suitable choice for everyday desktop use or for those seeking a quick and effortless setup. Instead, I strongly recommend starting with more user-friendly distributions such as Linux Mint or Ubuntu to build foundational confidence. Once comfortable with basic Linux commands and concepts, transitioning to a minimalist system like Kwort can provide an immensely rewarding next step for deepening system administration knowledge.

For learners specifically interested in advanced topics, Kwort (or similar CRUX-based distributions) serves as an excellent educational tool. It forces deliberate decision-making at every stage — from partitioning to package management — and cultivates a mindset of thoughtful customization and problem-solving. In academic or self-study contexts, combining Kwort installations with virtualization tools like Oracle VM VirtualBox is highly effective, as snapshots allow safe experimentation and easy recovery from mistakes.

In professional or development environments, I would recommend Kwort primarily for learning purposes, lightweight servers, or testing scenarios where minimal resource usage and a clean base are priorities. For production virtual machines requiring broad hardware compatibility and rapid deployment, more mainstream options with stronger virtualization support may be preferable.

In summary, my experience with Kwort Linux has been both challenging and enlightening. It has reinforced my belief that true mastery in computing often comes from engaging with systems that demand understanding rather than convenience. I look forward to building upon this foundation in future projects and encourage other students to embrace similar hands-on challenges. By balancing simplicity with deliberate effort, minimalist distributions like Kwort continue to offer timeless value in an increasingly complex technological landscape.

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